

Comparing the Effectiveness of Control Methodologies of a Hip-Knee-Ankle Exoskeleton During Squatting

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Abstract—Manual materials handling occupations often involve repetitive lifting, lowering, and carrying motions, which can lead to muscular fatigue and/or injury. The risk increases when loads must be worn on the body for the entirety of a job shift. Exoskeletons have been developed to assist these types of motions, but require the user to bear the weight of a load through their body. Load carriage exoskeletons have been developed to offload worn mass from the user to the ground through the device structure, but they have had limited success and have not been well studied in manual materials handling tasks. In this paper, we introduce a hip-knee-ankle exoskeleton and two control methods: virtual model control and gravity compensation. We compared the ability of each controller to reduce lower-limb muscle activity during squatting. Because the virtual model controller is tailored to squatting, we hypothesized that it would outperform gravity compensation. Both controllers were able to reduce the activity of major lower-limb muscle groups during squatting when compared to squatting with the exoskeleton turned off. Contrary to our original hypothesis, the gravity compensation controller generally outperformed the virtual model controller, which may have been caused by the gravity compensation controller having more consistent knee torque application and the virtual model controller requiring better per-user tuning and familiarization. These results indicate the efficacy of both controllers in reducing injury risk in the lower limbs during squatting.

I. INTRODUCTION

Occupations that require manual materials handling (MMH) are widespread across many diverse industries [1] and involve repetitive lifting, lowering, pushing, pulling, and carrying tasks in sometimes hazardous and unpredictable environments [2]. Bearing loads is liable to cause fatigue, which increases the risk of developing musculoskeletal disorders in the legs and lower back [3]–[5]. Furthermore, some occupations require continuous load bearing in the form of cumbersome personal protective equipment (PPE). While this PPE is necessary to prevent serious injury or death, it can further add to the daily loads workers experience [6], [7]. When combined with a worn load, lifting motions can easily lead workers to biomechanical fatigue. In MMH jobs, squatting is usually recommended as a safe lifting technique [8]. However, repetitive squatting while under load can still

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Fig. 1. The hip-knee-ankle exoskeleton used in this study was designed for load carriage of heavy personal protective equipment required for hazardous manual materials handling occupations. The device was tested in a human-participant experiment to validate the effectiveness of the virtual model and gravity compensation controllers in reducing muscular effort of the legs. The left image shows a front view of the device, while the right image shows a user squatting with an added air tank (not used in this paper).

be fatiguing and lead to the same injury risks induced by more traditionally dangerous postures [9]–[11].

Exoskeletons have the potential to address fatigue-induced injury caused by MMH tasks, such as load-bearing [12]. A majority of previous research has focused on devices designed for sagittal plane assistance during walking, usually assisting a single joint (hips, knees, or ankles) [13], [14]. Increasing interest from industry and research is being afforded to devices designed to prevent injuries caused by industrial MMH movements. These devices are often designed to provide assistive torque to upper-body biological joints either through passive elastic elements [15]–[17], or active actuation [18], [19], therefore reducing the risk of injury development. This method of reducing the effort of specific joints is similar to that of exoskeletons designed for walking.

However, while shown to be biomechanically beneficial

for specific movements and postures, contemporary MMH-centered exoskeletons require that the user bear the weight of all worn and lifted loads during operation. This proves problematic in settings where users must perform multiple MMH movements and transition between them, especially when loaded. Previous hip-knee-ankle exoskeletons have been designed specifically for this scenario by bypassing the human musculoskeletal system entirely and directly transferring the force of a worn load to the ground [20], [21]. While impressive in their potential to carry heavy loads, the devices were either cumbersome and detrimental to metabolic cost of walking or lacked evidence to prove efficacy [12].

Unsurprisingly, another factor responsible for the difficulty of developing load carriage exoskeletons is control complexity. These devices must compensate for extra worn loads while accounting for the mass and inertia of the device itself independent of user state or task (task-agnostic control). As a result, controllers that rely on accurate dynamic models of the structure and user force feedback have been implemented in load carriage exoskeletons [22]–[25]. However, tailoring assistance to a specific task (task-specific control) may be more beneficial to the user [26]. These previous devices and control methods have been studied in ambulation, but little research has been done applying these strategies to MMH movements [27].

Thus there exists a gap in the field for biomechanical evaluation of control methodologies for load carriage exoskeletons in MMH tasks. In this paper, we introduce a full lower-limb exoskeleton designed to offload heavy PPE required for hazardous MMH jobs along with two control strategies. The first is a virtual model controller [28] that outputs joint torques based on the compression/extension of a virtual linear spring placed along each leg and virtual revolute dampers about the hip and knee joints. The second is an inverse dynamics gravity compensation controller that applies torque based on the centers of pressure (CoP) of a user's feet and the device's dynamics, providing a pseudo-transparency mode to the user.

We compare the efficacy of both controllers in assisting squatting during a human-participant experiment (Fig. 1). The gravity compensation controller is similar to previous task-agnostic strategies used by other load carriage exoskeletons [22]–[25], while the virtual model controller is task-specific for squatting and utilizes virtual elements tuned and positioned specifically for squatting assistance. Therefore, we hypothesize that, during squatting, the virtual model controller will be more effective than the gravity compensation controller at reducing muscular effort in the legs.

II. METHODS

A. Exoskeleton Device

The exoskeleton used in this study (Fig. 1) is a custom device designed to mitigate the weight of a worn air tank and other heavy PPE by integrating directly with a self-contained breathing apparatus (SCBA) harness. The device is height adjustable using modular carbon fiber bars on the thighs and shanks, linking together custom metal

components that facilitate joint connections and rotation. Per leg, it features 5 actuated degrees of freedom (DoFs) (bilateral hip flexion/extension, knee flexion/extension, and ankle plantar flexion) and 4 passive DoFs at the hips: one revolute ab/adduction joint aligned with the user's biological hip joint, and a compliant 3D-printed hip linkage that allows for internal/external rotation. The exoskeleton, in total, is 19.3 kg.

The active DoFs consist of off-the-shelf brushless DC motors with in series 9:1 planetary gear reductions (T-Motor, Nanchang, Jiangxi, China). These are aligned with the hip and knee joints, while the ankles are remotely driven with the same actuators through Bowden cable systems and a 3:1 pulley reduction. The actuators feature integrated controllers using a controller area network (CAN) bus for torque, velocity, and position commands with input and output encoding.

The torso harness, thigh linkages, and shank linkages house custom F28388D (Texas Instruments, Dallas, Texas, USA) microcontroller boards with onboard 6-axis inertial measurement units (IMU). Signals input to these boards are output as EtherCAT on a daisy chain throughout the exoskeleton. The ankles are instrumented with in-line tension/compression load cells (Futek, Irvine, California, USA) and absolute encoders (Renishaw, Wotton-under-Edge, UK) for closed-loop tension control. The shoes are also instrumented with pressure-sensitive insoles (Noraxon USA, Scottsdale, AZ, USA), which are read using HTTP communication through Robot Operating System (ROS). Both the CAN signals and EtherCAT thread are controlled by an Intel NUC computer in real time (Simply NUC, Round Rock, Texas, USA) [29].

B. Virtual Model Controller

One of the high level control methods used to assist squatting in this study is a virtual model controller (Fig. 2.a) [28]. The controller employs virtual models of torsional dampers and linear springs to assist a user based on their squat depth, joint position, and squat velocity. The gains of the virtual springs and dampers are controlled by a finite state machine consisting of separate states for the downward and upward squatting motions.

For this study, unidirectional virtual torsional dampers were placed at the hip and knee joints. Virtual linear springs were placed along each leg, with endpoints at the hip and foot. These endpoints can translate a small distance (around 15 cm) horizontally to better align the spring force along the vector between the center of pressure and center of mass. The magnitude of force generated by each linear spring F_s is given by

$$F_s = -k\Delta x \quad (1),$$

in which k is the virtual spring constant and Δx is the spring's deflection from its resting position. The force exerted from each spring on the exoskeleton can be expressed as an external wrench $\mathcal{F}_s$ applied at the exoskeleton feet with

$$\mathcal{F}_s = [0 \ 0 \ 0 \ -F_{sx} \ -F_{sy} \ -F_{sz}]^T \quad (2),$$

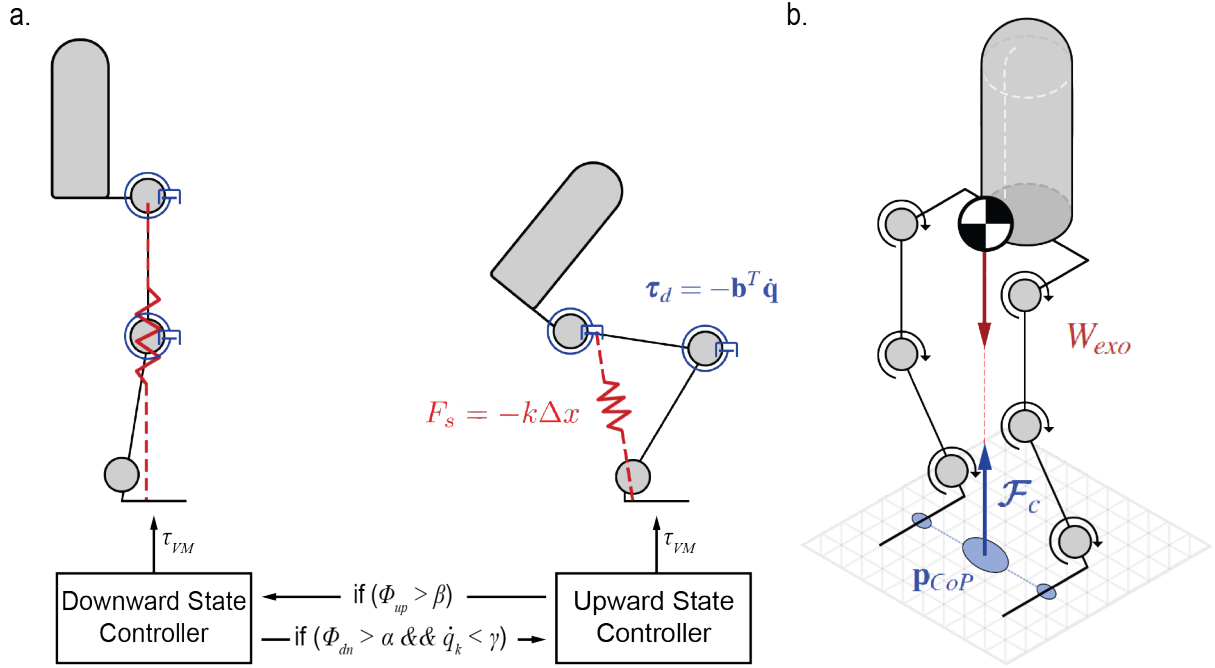


Fig. 2. The exoskeleton is controlled by either of two high-level controllers. The virtual model controller (a) consists of torsional dampers placed at the hip and knee joints in flexion/extension, and a virtual linear spring pinned between the hip and the foot. The connection points of the linear spring have the ability to translate horizontally a small fixed distance to better align the spring force along the vector between the center of pressure and center of mass. The controller operates using a finite state machine comprised of two sub-states. The downward state occurs when the user is standing or descending, during which the virtual dampers and linear spring are both active. The upward state occurs when the user is ascending, during which the dampers are deactivated, and the stiffness of the spring is increased to provide additional assistance during the upward motion. The transition to the upward motion controller occurs when the downward controller's phase variable ϕ_{dn} crosses α , and the knee angular joint velocity $\dot{q}_k$ drops below γ . The transition back to the downward motion controller occurs when the upward controller's phase variable ϕ_{up} crosses β . The phase variables for each controller are based on current knee angle relative to absolute minimum and maximum knee angles when squatting. α , β , and γ are all tunable. Conversely, the gravity compensation controller (b) operates agnostic of user state, and uses a model of the exoskeleton's mass and inertia properties as well as the centers of pressure of the feet (sensed by insoles). At each joint configuration, torque is applied to the joints to compensate for the exoskeleton's dynamics.

where the terms F_{sx} , F_{sy} , and F_{sz} represent the x , y , and z -direction components of the spring force expressed in the wrench frame. The wrench is then mapped to the vector of desired sagittal hip, knee, and ankle flexion/extension joint torques, τ_s , using the transpose of the exoskeleton leg jacobian, $\mathbf{J}(\mathbf{q})^T$ as

$$\tau_s = \mathbf{J}(\mathbf{q})^T \mathcal{F}_s \quad (3),$$

where $\mathbf{q} = [\mathbf{q}_0, \mathbf{q}_n] \in \mathbb{R}^{6+n}$, and where $\mathbf{q}_0 \in \mathbb{R}^6$ is the floating base and $\mathbf{q}_n \in \mathbb{R}^n$ is the vector of joint angles for an exoskeleton with n joints. The torque generated by the virtual torsional dampers (τ_d) upon squat descent is given by

$$\tau_d = -\mathbf{b}^T \dot{\mathbf{q}} \quad (4),$$

using the virtual damping coefficients, $\mathbf{b}$, and the angular flexion/extension velocity of the joints, $\dot{\mathbf{q}}$.

The total desired torque sent to the actuators including scaling factor $\sigma_{scale} \in [0, 1]$ can then be expressed as

$$\tau_{VM} = \sigma_{scale} [\mathbf{J}(\mathbf{q})^T \mathcal{F}_s - \mathbf{b}^T \dot{\mathbf{q}}] \quad (5).$$

During downward squatting motion, spring stiffness is set low and damping is set relatively high to control the

descent of the user. During the upwards squatting motion, damping is removed, and the stiffness on the linear springs is instantaneously increased, to give the user a boost upwards. Several preliminary user tests were conducted to determine a comfortable ratio of k relative to $\mathbf{b}$ for both directions of motion. This ratio was then maintained, while the overall magnitude of resulting assistance was tuned using σ_{scale} per user preference. The phase variables for each sub-controller (ϕ_{dn} , ϕ_{up}) are based on current knee angle relative to absolute minimum and maximum knee angles when squatting. The transition from the downward motion controller to the upward motion controller occurs when the downward controller's phase variable ϕ_{dn} crosses a chosen threshold α while descending, and the knee joint's angular velocity $\dot{q}_k$ drops below a chosen threshold γ (bottom of the squat). The transition back to the downward motion controller occurs when the upward controller's phase variable ϕ_{up} crosses a chosen threshold β while ascending (fully standing). The thresholds α , β , and γ can all be tuned per user preference.

C. Inverse Dynamics Gravity Compensation Controller

While assistance of the virtual model controller is dependent on direction and speed of squatting motions, gravity

compensation is designed to be movement agnostic and support the dynamics of the exoskeleton and any added weight and inertia to the human-exoskeleton system (Fig. 2.b). An accurate dynamic model of the exoskeleton is used, as well as pressure-sensitive insoles to sense the CoPs of the user's feet. The joint space dynamics of the exoskeleton are written as

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\tau} \end{bmatrix} + \mathbf{J}(\mathbf{q})^T \mathcal{F}_c \quad (6),$$

where $\mathbf{M}(\mathbf{q})$ is the exoskeleton's mass and inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are the vectorized Coriolis and centrifugal forces at each joint, $\mathbf{G}(\mathbf{q})$ are the vectorized torques resulting from the gravitational forces acting on each linkage of the exoskeleton, $\boldsymbol{\tau}$ are the input torques, $\mathcal{F}_c$ are the concatenated ground reaction wrenches at the feet, and $\mathbf{J}(\mathbf{q})^T$ are the concatenated leg Jacobians.

Flexion/extension joint torques are calculated from the estimated ground reaction forces on each leg as

$$\boldsymbol{\tau}_{\text{comp}} = \mathbf{J}(\mathbf{q})^T \hat{\mathcal{F}}_c + \mathbf{G}(\mathbf{q}) \quad (7),$$

where $\hat{\mathcal{F}}_c$ is the estimated ground reaction wrench generated by the exoskeleton, such that the torques compensate for the weight of the exoskeleton and the exoskeleton's estimated ground reaction forces.

One of the challenges is that the ground reaction wrench is not directly observable. Therefore, we estimate the ground reaction wrench of each foot using information from the insoles. The insoles report a fraction of total ground reaction force α and a center of pressure $\mathbf{p}_{CoP} \in \mathbb{R}^3$ for each insole. The normal force at the foot is estimated as

$$\hat{\mathbf{f}}_c = [0, 0, \alpha W_{\text{exo}}]^T \quad (8),$$

where W_{exo} is the weight of the exoskeleton. Each foot's ground reaction wrench is then estimated as

$$\hat{\mathcal{F}}_c = [\hat{\mathbf{f}}_c, \mathbf{p}_{CoP} \times \hat{\mathbf{f}}_c]^T \quad (9).$$

Before operation, the percentage of weight compensated for by the device is tuned per user as to not impede the user's squatting motions. This results in the desired torque sent to the motors being defined as

$$\boldsymbol{\tau}_d = k_{\text{scale}} \boldsymbol{\tau}_{\text{comp}} \quad (10),$$

where $k_{\text{scale}} \in [0, 1]$ is this percentage of weight gain. Exoskeleton assistance begins to inhibit user motion as the percentage of offloaded weight nears 100%. We hypothesize this is due to discrepancies between the modeled exoskeleton and actual exoskeleton as well as slight inaccuracies in ground reaction wrench estimations.

D. Validation Experiment

To compare the effectiveness of the virtual model and gravity compensation controllers in assisting users during squatting motions, we recruited five participants (average age of 26.2 years, mass of 72.9 kg, and height of 169.5

cm, two female, three male) for a validation experiment. This was approved by the Florida Institute for Human and Machine Cognition Institutional Review Board (IRB-2022-0042). Before participation, each subject provided informed consent in accordance with institute standards and confirmed they had no prior history of lower-body musculoskeletal injury or any other biomechanical issues that would confound the results of experimentation.

All data collections took place in the experimental setup as shown in (Fig. 3). Subjects squatted with four conditions: NO EXO (without the exoskeleton), EXO OFF (exoskeleton equipped but powered off), GC (exoskeleton is powered on and assisting using gravity compensation), and VMC (exoskeleton is powered on and assisting using virtual model control). To mitigate against ordering effects, conditions were completed in a double reversal pattern of ABCDDCBA (A = NO EXO, B = EXO OFF, C and D = GC or VMC), resulting in eight total data collections. The C and D conditions were randomized.

Under all conditions, subjects stood with their feet placed shoulder-width apart and squatted, bending primarily at the hips and knees, until their gluteus muscles contacted a bench 50 cm off the ground. Once this occurred, the subjects were instructed to immediately begin ascending, as to prevent instantaneous sitting on the bench. Squatting speed was controlled by a metronome ticking at 90 beats per minute with two ticks each for ascent and descent timing. Subjects squatted for 30 seconds for each condition, resulting in around 12 repetitions. Between every condition, subjects rested for at least 60 seconds, and a five minute rest period was enforced after the first D condition.

During a familiarization period immediately prior to experimentation, subjects were instructed on proper squatting technique without the exoskeleton and were asked to do squatting repetitions with the metronome for no more than 30 seconds at a time. After participants became comfortable, they donned the exoskeleton and were asked to complete more repetitions until comfort was achieved again. After this,

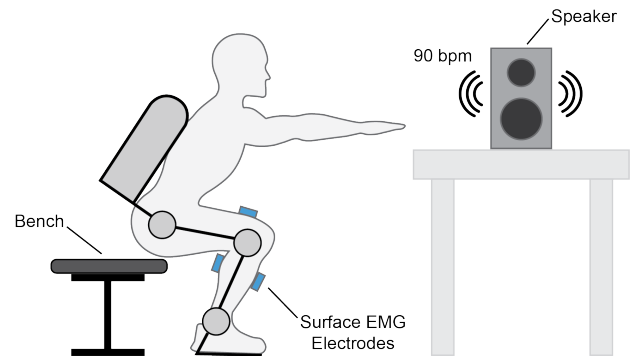


Fig. 3. The experimental setup. Participants were instructed to stand with their feet shoulder-width apart, squat down until they touch the bench with their gluteus muscles, and ascend until standing. The pace of squatting was controlled using a speaker playing a metronome at 90 beats per minute, taking 2 beats each to descend and ascend. Electromyography data were collected with surface electrodes (pictured in blue) throughout each condition.

the magnitude of virtual model controller assistance (σ_{scale}) and the percentage of exoskeleton weight offloaded by the gravity compensation controller (k_{scale}) were tuned (both on a scale of 0 to 100%). Specifically for the virtual model controller, the thresholds α and γ were kept constant, and β was tuned per user. For each controller, subjects were asked to perform squats while the magnitudes were being changed. As soon as participants found the assistance inhibited motion, parameters were decreased slightly, tuning ceased, and the participant doffed the exoskeleton.

During each experiment, electromyography (EMG) data from left and right leg rectus femoris (RF), vastus medialis (VM), gastrocnemius lateralis (GL), and tibialis anterior (TA) groups were collected. This was done using an Ultium system from Noraxon (Noraxon, Scottsdale, Arizona, USA) at 1000 Hz. Electrodes were placed according to SENIAM guidelines on each of the measured muscle groups [30]. Prior to collection, placements were verified by experimenters by asking the subjects to activate each muscle group and visually inspecting the measured signal for motion artifact or cross talk. The EMG data collected in the experiment were high-pass filtered at 20 Hz, rectified, then low-pass filtered at 6 Hz to calculate the linear envelope [31]. The root-mean-square (rms) of the second half of each condition was calculated to remove transient effects as participants became accustomed to the squatting rhythm. The root-mean-squared EMG results of each muscle group and condition were averaged across the five participants. After averaging, the EXO OFF, GC, and VMC conditions were normalized to the NO EXO condition.

III. RESULTS

Throughout the experiments, the gravity compensation and virtual model controllers generated torque patterns that were comfortable and assisted the participants in squatting (Fig. 4). Both controllers were effective in reducing the average root-mean-squared EMG (rmsEMG) activation of each muscle group relative to the EXO OFF condition in the validation experiment (Fig. 5). The muscle activation changes of each condition are shown, while the NO EXO condition is represented by the dotted line. The rmsEMG of each muscle increased during the EXO OFF, some by over 50% (right and left RF, left TA).

Relative to the EXO OFF condition, both the gravity compensation and virtual model controllers reduced activity across all muscles. The thigh muscles (RF and VM) experienced the greatest activity reductions, (-32% on left RF), while the shank muscles (TA and GL) experienced generally lower magnitudes of activity reduction (-15.2% to -8.2% on GL), no change in activity, or a slight increase in activity (from -0.4% to +6.4% on TA). RF experienced the largest rmsEMG reductions and TA experienced the smallest rmsEMG reduction relative to the EXO OFF condition. In fact, TA muscle activity increased with the virtual model controller.

Across the muscle groups, the gravity compensation controller was more effective in activity reduction than the virtual model controller. The difference between the two

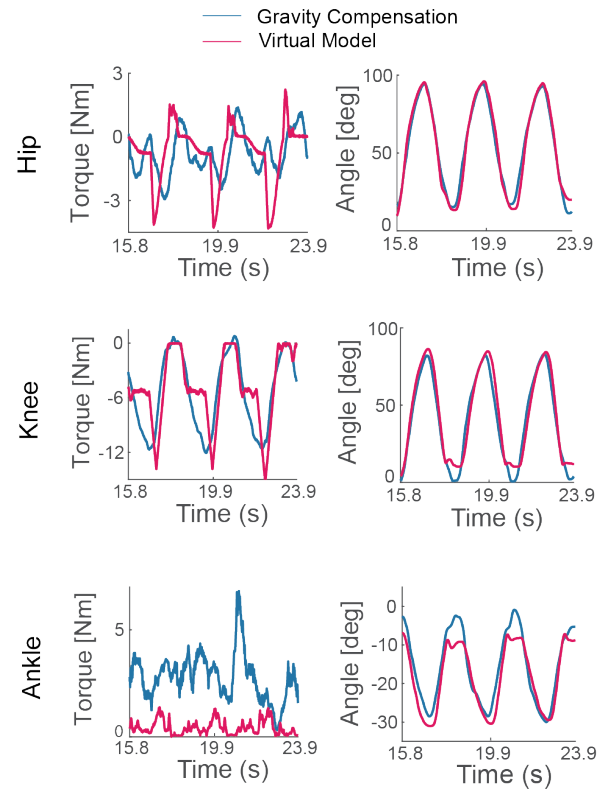


Fig. 4. Representative joint-level torque values generated from the controllers with the corresponding joint angles over three squat repetitions. Gravity compensation is shown in blue and virtual model is shown in red. From top to bottom, the plots show the hip (extension +, flexion -), knee (flexion +, extension -) and ankle (plantarflexion +) torque (left) and joint angle (right). The joint angles 0 when standing straight.

was the greatest in the RF group (a difference of 12.9% on left RF). However, the controllers performed similarly considering the VM and GL groups (a difference of 3.1% on right VM and 7.4% on right GL, and a difference of 2.0% on left VM and 1.4% on left GL).

IV. DISCUSSION

The results show that both controllers were able to reduce the amount of muscle activity required to squat while the exoskeleton was equipped compared to the EXO OFF condition. In some cases, muscle activity was decreased to near the NO EXO activity level, especially in the thigh muscles. The shank muscle rmsEMG results also show decreases from EXO OFF, but are generally more variable. This could be due to the overall activity in the TA and GL muscles being low across the conditions, causing signal noise to be more prevalent in the data. Reducing muscle activity to close to the NO EXO condition was expected for the gravity compensation controller as it was designed to counter the dynamics of the device.

We hypothesized that the virtual model controller would perform better than gravity compensation, due to it being specifically designed for squatting. However, the gravity compensation controller tended to perform slightly better

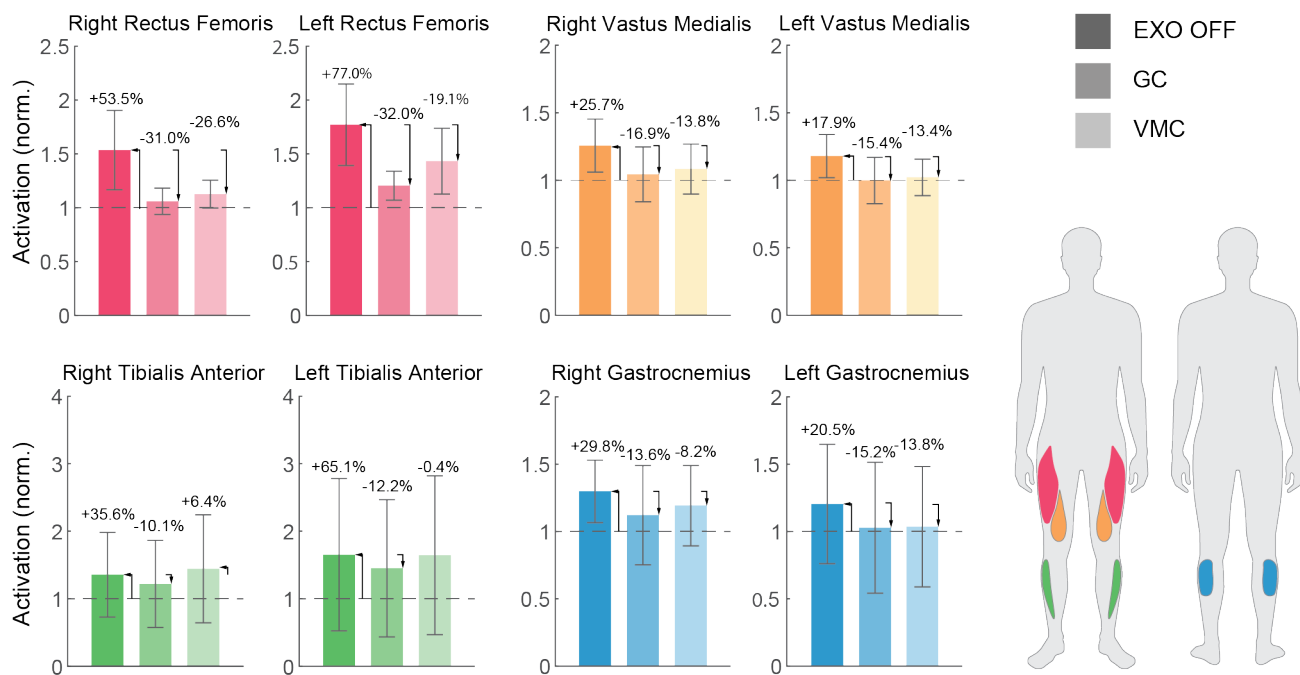


Fig. 5. The rmsEMG data for each muscle group are shown. Each condition is normalized to the NO EXO condition (represented by the dotted line). The percentages represent change in rmsEMG of (1) EXO OFF relative to NO EXO and (2) GC and VMC relative to EXO OFF. The bilateral RF are red, VM are orange, TA are green, and GL are blue. Within each muscle group, the experimental conditions are represented by different color saturation.

than the virtual model controller, contradicting our hypothesis. This result is likely in part due to the necessity for a longer familiarization period. The virtual model controller can provide quite different assistance experiences based on how the user operates the device and adjusts their squatting technique. We found that many users were trying to resist the assistance when using this control mode in order to maintain the squatting cadence required by the experimental design. The difference in performance between the controllers could also be due to the differences in torque application, specifically at the knees. As seen in (Fig. 4), the torque patterns of the controllers are quite different: where the gravity compensation profile is smoother and close to sinusoidal, and the virtual model controller profile consists of a large jump in torque at the onset of upward motion, then a sharp decrease to zero. The transition from descent to ascent is the highest-activation portion of squatting [32]. The differences in rmsEMG reductions could be because the gravity compensation controller applies more consistent knee torque than the virtual model controller before the onset of upward motion. These shortcomings of the virtual model controller could be mitigated by implementing a more sophisticated tuning process for its parameters [26], and employing an improved training protocol.

While the device was able to reduce user muscle activity, there were some limitations to this study. One limitation is that ankle torque was low for each controller. In the virtual model controller, this can be explained by the virtual spring ankle connection being close to the ankle joint itself, regardless of the horizontal degree of freedom. Because there

is very little distance from the joint to the connection point, little torque was generated. In the gravity compensation controller, by the structure of the exoskeleton system itself, little ankle torque was necessary to maintain posture. Ankle torque alone can have a positive effect on biomechanical outcomes [19]. Therefore, larger ankle torque magnitudes could have improved the performance of the exoskeleton.

Another limitation is that the experiment was conducted without adding any extra mass on the exoskeleton limiting the biomechanical outcomes relative to the NO EXO condition. For example, the gravity compensation controller was designed to both counter the dynamics of the exoskeleton itself as well as extra worn mass. Added weight could have made the effects of the virtual model controller more apparent as well. However, the purpose of the experiment described in this paper was to determine baseline differences between the two controllers.

Future work should investigate the efficacy of these controllers during loaded squatting representative of MMH conditions across more subjects. Combining the two controllers may produce greater biomechanical improvements than either controller separately, as gravity compensation can assist load carriage and the virtual model controller can provide task specific assistance. Given that gravity compensation is a task-agnostic controller, future experiments should explore the effect of gravity compensation on a variety of MMH-relevant tasks including but not limited to squatting. Additionally, we intend to explore more sophisticated approaches to tuning the virtual model controller to obtain optimal parameters for specific individuals and general users.

V. CONCLUSION

This paper presented two methods of controlling a full lower-body exoskeleton during squatting. While both were designed to provide assistive joint torques throughout the RoM of a squat, they do so in distinct ways. The virtual model controller uses virtual mechanical elements to apply a linear spring force resolved as hip, knee and ankle torques. Conversely, the gravity compensation controller utilizes a dynamic model of the exoskeleton and estimated foot ground reaction forces to compute the inverse dynamics of a particular pose, compensating for the dynamics of each linkage.

We conducted an experiment to assess the effectiveness of both controllers during squatting motions compared to NO EXO and EXO OFF conditions. We observed reductions in rmsEMG in the user's legs in both controllers relative to the EXO OFF condition, and thigh muscle rmsEMG was nearly reduced to NO EXO levels. Gravity compensation appeared to be more effective at reducing muscle activity during squatting than the virtual model controller. This is possibly due to the gravity compensation controller producing smoother knee torque profiles that better matched the activation of the user's muscles and the virtual model controller requiring more time for familiarization and more specific tuning.

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